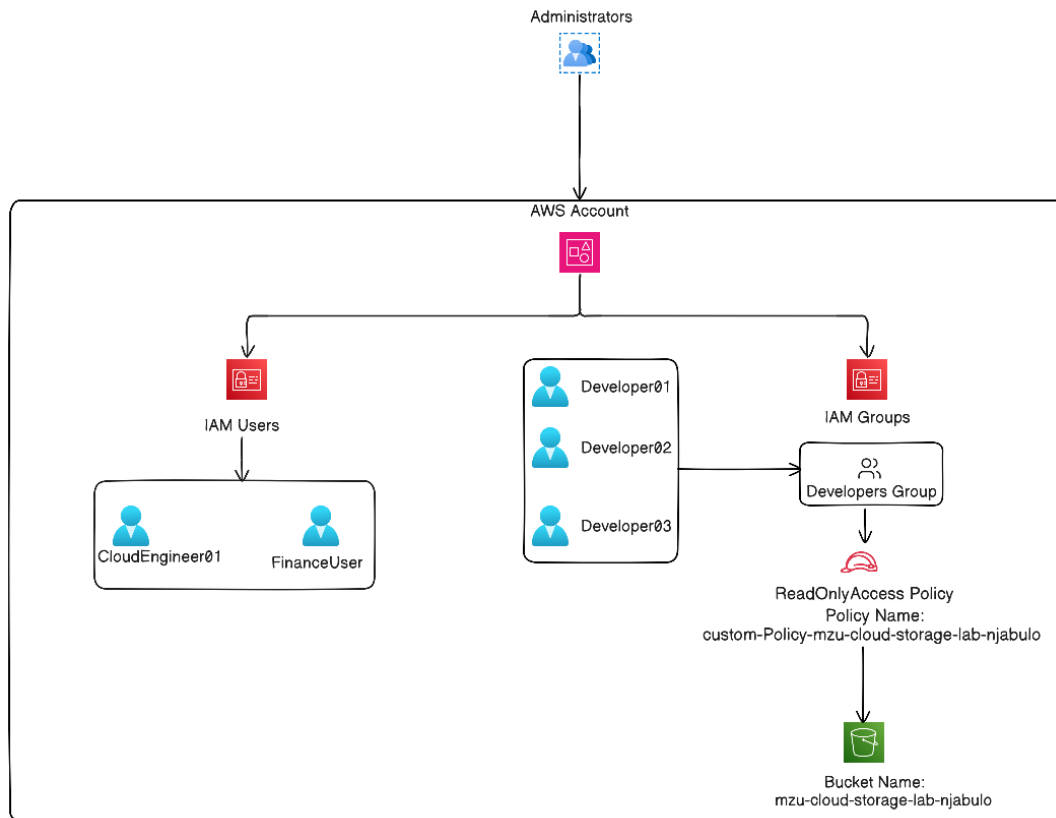


Identity & Access Management (IAM)

LAB Architecture



Task 1 - Navigate to IAM

The screenshot shows the AWS IAM Dashboard. The left sidebar contains navigation links for **Identity and Access Management (IAM)**, **Dashboard**, **Access Management** (Roles, Policies, IAM users, IAM user groups, Identity providers, Account settings, Root access management, Temporary delegation requests), **Access reports** (Access analyzer, Resource analyzer, Unused access, Analyzer settings, Policy simulator, Credential report, Organization activity, Service control policies, Resource control policies), and **Related console** (IAM Identity Center, AWS Organizations). The main content area displays **Security recommendations** (Root user has MFA, Root user has no active access keys), **IAM resources** (Resources in this AWS Account: User groups: 2, Users: 2, Roles: 4, Policies: 0, Identity providers: 0), and **What's new** (Updates for features in IAM). The right sidebar shows **AWS Account** information (Account ID: 208073622332, Account Alias: Account Alias, Create, Sign-in URL for IAM users in this account: https://208073622332.signin.aws.amazon.com/console) and **Quick Links** (My security credentials, Manage your access keys, multi-factor authentication (MFA) and other credentials). The bottom right section includes **Tools** (Policy simulator) and **Additional information** (Security best practices in IAM, IAM documentation, Videos, blog posts, and additional resources).

Task 2 - Create an IAM User

Username:

- **CloudEngineer01**

IAM users (1/3) [Info](#)

Q Search

User name

Path

Group

Last activity

MFA

Password age

Console last sign-in

Access key ID

Active key age

Access key last use

ARN

Creation time

Consol

☒	CloudEngineer01	/	0	-	-	1 minute	-	-	-	-	arn:aws:iam:208073622332:user/Clou...	1 minute ago	Enal
-------------------------------------	-----------------	---	---	---	---	----------	---	---	---	---	---------------------------------------	--------------	------

⌂

Delete

Create user

Task 3 - Create a Second User

Username

- **FinanceUser01**

IAM users (1/4) [Info](#)

Q Search

User name

Path

Group

Last activity

MFA

Password age

Console last sign-in

Access key ID

Active key age

Access key last use

ARN

Creation time

Consol

☐	CloudEngineer01	/	0	-	-	6 minutes	-	-	-	-	arn:aws:iam:208073622332:user/Clou...	6 minutes ago	Enal
☒	FinanceUser01	/	0	-	-	1 minute	-	-	-	-	arn:aws:iam:208073622332:user/Fina...	1 minute ago	Enal

⌂

Delete

Create user

Task 4 - Create an IAM Group

Group Name

- **Developers**

Attach the AWS Managed Policy

- **ReadOnlyAccess**

Developers [Info](#)

⌂

Delete

Summary

User group name

Developers

Creation time

July 30, 2026, 20:49 (UTC+02:00)

ARN

arn:aws:iam:208073622332:group/Developers

⌂

Edit

Users

Permissions

Access Advisor

Permissions policies (1) [Info](#)

⌂

Simulate [i](#)

Remove

Add permissions

Q Search

Filter by Type

All types

Policy name

Type

Attached entities

☐	IAMReadOnlyAccess	AWS managed	1
--------------------------	-----------------------------------	-------------	---

Task 5 – Add Users to the Group

Open

- **Developers Group**

Choose

- Add Users

Select

- **CloudEngineer01**

Save

developers info Delete

Summary Edit

User group name Developers	Creation time July 30, 2026, 20:49 (UTC+02:00)	ARN arn:aws:iam::208073622332:group/Developers
-------------------------------	---	---

[Users \(1\)](#) | [Permissions](#) | [Access Advisor](#)

Users in this group (1) Remove Add users

An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.

Q Search

☐	User name ↗	Groups	Last activity	Creation time
☐	CloudEngineer01	1	None	15 minutes ago

Verify

- *CloudEngineer01 has **ReadOnlyAccess** permission*

CloudEngineer01 info Delete

Summary

ARN arn:aws:iam::208073622332:user/CloudEngineer01	Console access Enabled without MFA	Access key 1 Create access key
Created July 30, 2026, 20:37 (UTC+02:00)	Last console sign-in Never	

[Permissions](#) | [Groups \(1\)](#) | [Tags](#) | [Security credentials](#) | [Last Accessed](#)

Permissions policies (1) Simulate ↗ Remove Add permissions ▼

Permissions are defined by policies attached to the user directly or through groups.

Q Search

Filter by Type All types

☐	Policy name ↗	Type	Attached via ↗
☐	IAMReadOnlyAccess	AWS managed	Group Developers

Task 6 – Review AWS Managed Policies

Task 7 – Create a Custom Policy (Optional)

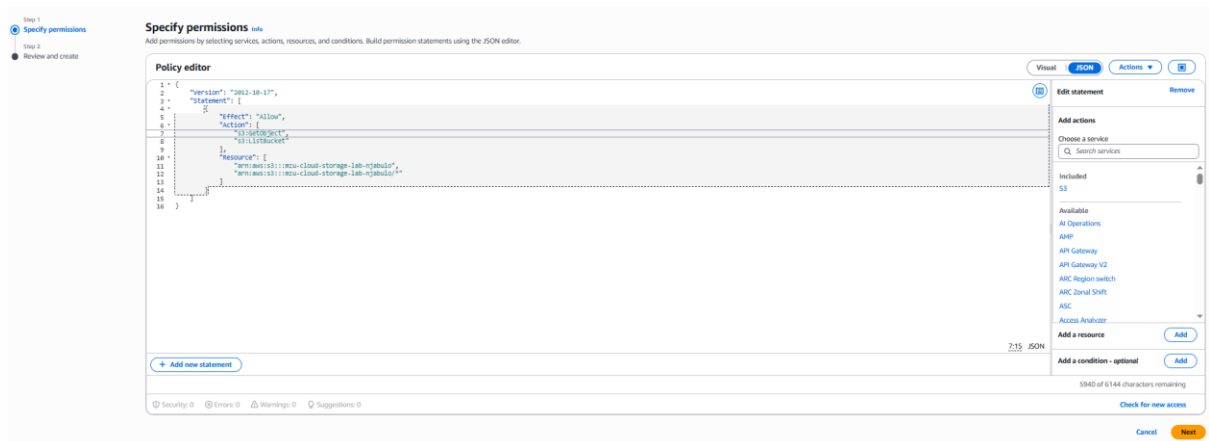
Custom-managed Policy that allow readd-only access to a single S3

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
```

```

"Action": [
    "s3:GetObject",
    "s3:ListBucket"
],
"Resource": [
    "arn:aws:s3:::mzu-cloud-storage-lab-njabulo",
    "arn:aws:s3:::mzu-cloud-storage-lab-njabulo/*"
]
}
}
}

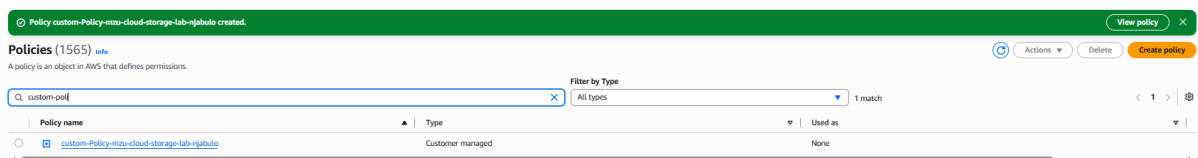
```



Customer Policy name

- **custom-Policy-mzu-cloud-storage-lab-njabulo**

Policy Created



Task 8 - Enable Multi-Factor Authentication

User

- CloudEngineer01

CloudEngineer01

Info

Delete

Summary

ARN

arn:aws:iam::208073622332:user/CloudEngineer01

Console access

Enabled with MFA

Access key 1

Create access key

Created

July 30, 2026, 20:37 (UTC+02:00)

Last console sign-in

Never

Task 9 – Create an IAM Role

IAM

Roles

Create role

Step 1: Select trusted entity

Step 2: Add permissions

Step 3: Name, review, and create

Select trusted entity

Trusted entity type

☒ AWS service

☐ AWS account

☐ Web identity

☐ SAML 2.0 federation

☐ Custom trust policy

Use case

Service or use case

EC2

Choose a use case for the specified service.

Use case

☒ EC2

☐ EC2 Role for AWS Systems Manager

☐ EC2 Spot Fleet Role

☐ EC2 - Spot Fleet Auto Scaling

☐ EC2 - Spot Fleet Tagging

☐ EC2 - Spot Instances

☐ EC2 - Spot Fleet

☐ EC2 - Scheduled Instances

Attach the AWS Managed Policy

- AmazonS3ReadOnlyAccess

Role Name

- EC2-S3-ReadOnlyAccess-Role

Roles (1/5)

Info

Delete

Create role

Search

Q

Role name	Trusted entities	Last activity
AWSManagedPolicyForSSM	AWS Service: rds (Service-Linked Rol	5 days ago
AWSManagedPolicyForResourceExplorer	AWS Service: resource-explorer-2 (S	43 minutes ago
AWSManagedPolicyForSupport	AWS Service: support (Service-Link	-
AWSManagedPolicyForTrustedAdvisor	AWS Service: trustedadvisor (Servic	-
☒ EC2-S3-ReadOnlyAccess-Role	AWS Service: ec2	-

Assigned CloudEngineer01

- custom-Policy-mzu-cloud-storage-lab-njabulo

CloudEngineer01 Info

Summary

ARN

arn:aws:iam::208073622332:user/CloudEngineer01

Created

July 30, 2026, 20:37 (UTC+02:00)

Console access

Enabled with MFA

Last console sign-in

11 days ago

Access key 1

[Create access key](#)

Permissions

Groups (1)

Tags

Security credentials

Last Accessed

Permissions policies (4)

Simulate

Remove

Add permissions

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type

All types

Search

1

☐	Policy name	Type	Attached via
☐	AmazonEC2ReadOnlyAccess	AWS managed	Directly
☐	AmazonS3ReadOnlyAccess	AWS managed	Directly
☐	custom-Policy-mzu-cloud-storage-lab-njabulo	Customer managed	Directly
☐	IAMReadOnlyAccess	AWS managed	Group Developers

Task 10 – Permission Investigation

Question 1

- Can CloudEngineer01 delete an S3 bucket? No
- User not granted rights to delete or modify the S3/objects etc ect

Amazon S3

Failed to delete objects

For more information, see the Error column in the Failed to delete table below.

Diagnose with Amazon Q

Delete objects: status

Close

After you navigate away from this page, the following information is no longer available.

Summary

Source

s3://mzu-cloud-storage-lab-njabulo

Successfully deleted

0 objects

Failed to delete

1 object

Failed to delete

Configuration

Failed to delete (1 object, 0 B)

Find objects by name

Name	Folder	Type	Last modified	Size	Error
architecture-plan.txt	-	txt	July 30, 2026, 15:27:59 (UTC+02:00)	0 B	Access denied

Delete bucket Info

This bucket is not empty

Buckets must be empty before they can be deleted.

Empty bucket

Diagnose with Amazon Q

Delete bucket "mzu-cloud-storage-lab-njabulo"?

To confirm deletion, enter the name of the bucket in the text input field.

mzu-cloud-storage-lab-njabulo

Cancel Delete bucket

Question 2

Can CloudEngineer01 view EC2 instances?

Answer: No – if the instance is stopped CloudEngineer01 can not start it.

Failed to start the instance i-08d8da9cd4d5e889

You are not authorized to perform this operation. User: arn:aws:iam::208073622332:user/CloudEngineer01 is not authorized to perform: ec2:StartInstances on resource: arn:aws:ec2:us-east-1:208073622332:instance/i-08d8da9cd4d5e889 because no identity-based policy allows the ec2:StartInstances action. Encoded authorization failure message: zuMzV6-okdaMTE-6Sdy5tESTs8tLzmH4MhMqyjlJ5ZiHfYomwswQpdp0i4dUurc_gbSd5v8mMPvWpJwWkImu5r-dGjYu0a00f8pzcOm6mGtG4t1X1uZePJQ9ANhJlcu9T1AkSqV9wWqVO4GycNwm8BOe-rnz5vJ8P1LNBbbl6v6eydDAFNdeBVJCQ1yGKNwGczJec17H4UPM487gLUKX6rPvS5Vofe42J7CM4jY1eKZyZaGb0R83Z-71madj2hVQWvBRNZCDdSRWjgwp2dal7mnX3VexUJQJjCylJueuM0zg175ye1e6ndXSK3q8_UvVteJ_hKaVxzhG5SRkZLsdMnjilBLzFHF_IK8eIkQqwkSLT3y7HcvMuXWmwTdxPGzhghIKOgtoHG8KO10qcx0Tnu9czpMLJBuzZPS7_Lre7elabiWnQDIEvOpQa0FNvH9a1L14tvtvSns1upplJ5nQ4fg61fm6DbaUhfBoCXaH_CzNHZeNcyWV3qu0LtcRIS6k1Y34Yy0Bq_5gfFwb7un51HKVOpDFSHOfuVahmkEy-x2XJ0DpwwdWky7RkocZhUTVxuugdwB9lTGKm5AAOznjghrJHigU2O4mzERtvCa8osv8BR5plU9JKA7XvtrwqfggczOxuxLulFVrmy4oQyCyuruR0aiPE6w3CsueBc8ax7Pbk7O4gJCFvB1FWl8p087Lfta9hXLZukuzp2Rq3DPQJ6Nic6Z9F-9UQc0Xqu8Qe5t3avWU4V021eRUG7ahQ2w-tpHogru5sWuqDQ00_GWcwFOYfCKNZsd0KcPYL6C7CDzeyvY9lpDJ6FDCQRPACHgNOPfzkg3h-gxqom2

Diagnose with Amazon Q

Instances (1/1)

info

Last updated 1 minute ago

Connect

Instance state

Actions

Launch instances

Save filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

Name

Instance ID

Instance state

Instance type

Status check

Availability Zone

Public IPv4 DNS

Public IPv4 ...

Elastic IP

IPv6 IPs

Monitorin

MyFirstEC2

i-08d8da9cd4d5e889

Stopped

t3.micro

-

us-east-1d

-

-

-

disabled

Challenge Lab

Create Users

- **Developer01**
- **Developer02**
- **Developer03**

IAM users (3/7)

info

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Search

	User name	Path	Groups	Last activity	MFA	Password age	Console last sign-in	Access key ID	Active key age	Access key last use	ARN	Creation time	Cor
☐	CloudEngineer01	/	1	27 minutes ago	Virtual...	34 minutes	37 minutes ago	-	-	-	arn:aws:iam::208073622332:user/CloudEngineer01	2 hours ago	☐
☒	Developer01	/	1	-	-	2 minutes	-	-	-	-	arn:aws:iam::208073622332:user/Developer01	2 minutes ago	☐
☒	Developer02	/	1	-	-	1 minute	-	-	-	-	arn:aws:iam::208073622332:user/Developer02	1 minute ago	☐
☒	Developer03	/	1	-	-	1 minute	-	-	-	-	arn:aws:iam::208073622332:user/Developer03	1 minute ago	☐
☐	FinanceUser01	/	0	-	-	2 hours	-	-	-	-	arn:aws:iam::208073622332:user/FinanceUser01	2 hours ago	☐
☐	njam	/	0	20 days ago	Virtual...	23 days	20 days ago	-	-	-	arn:aws:iam::208073622332:user/njam	23 days ago	☐
☐	student01	/	1	3 days ago	Virtual...	9 days	9 days ago	-	-	-	arn:aws:iam::208073622332:user/student01	9 days ago	☐

Developers

info

Delete

Summary

Edit

User group name

Developers

Creation time

July 30, 2026, 20:49 (UTC+02:00)

ARN

arn:aws:iam::208073622332:group/Developers

Users (4)

Permissions

Access Advisor

Users in this group (4)

Remove

Add users

An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.

Search

	User name	Groups	Last activity	Creation time
☐	CloudEngineer01	1	39 minutes ago	2 hours ago
☐	Developer01	1	None	5 minutes ago
☐	Developer02	1	None	3 minutes ago
☐	Developer03	1	None	3 minutes ago

Assign - all dev users

- **Developers (ReadOnlyAccess)**

Developers
info
Delete

Summary

User group name

Developers

Creation time

July 30, 2026, 20:49 (UTC+02:00)

ARN

am:aws:iam:208073622332:group/Developers

Edit

Users (4)

Permissions

Access Advisor

Permissions policies (1) info

Simulate i2

Remove

Add permissions

You can attach up to 10 managed policies.

Search

Filter by Type

All types

< 1 >

☐	Policy name ↗	Type	Attached entities
☐	IAMReadOnlyAccess	AWS managed	1

Bonus Challenge

Create IAM Group

- **Administrators**

Assign

- **AdministratorAccess**

Administrators
info
Delete

Summary

User group name

Administrators

Creation time

July 30, 2026, 23:26 (UTC+02:00)

ARN

am:aws:iam:208073622332:group/Administrators

Edit

Users

Permissions

Access Advisor

Permissions policies (1) info

Simulate i2

Remove

Add permissions

You can attach up to 10 managed policies.

Search

Filter by Type

All types

< 1 >

☐	Policy name ↗	Type	Attached entities
☐	AdministratorAccess	AWS managed - job function	2

Discussion: small number of users are granted admin-access

- Only trained users are granted admin access to perform admin related e.g. creating user profile and to prevent accidental changes and making it easy to track activity/ actions.